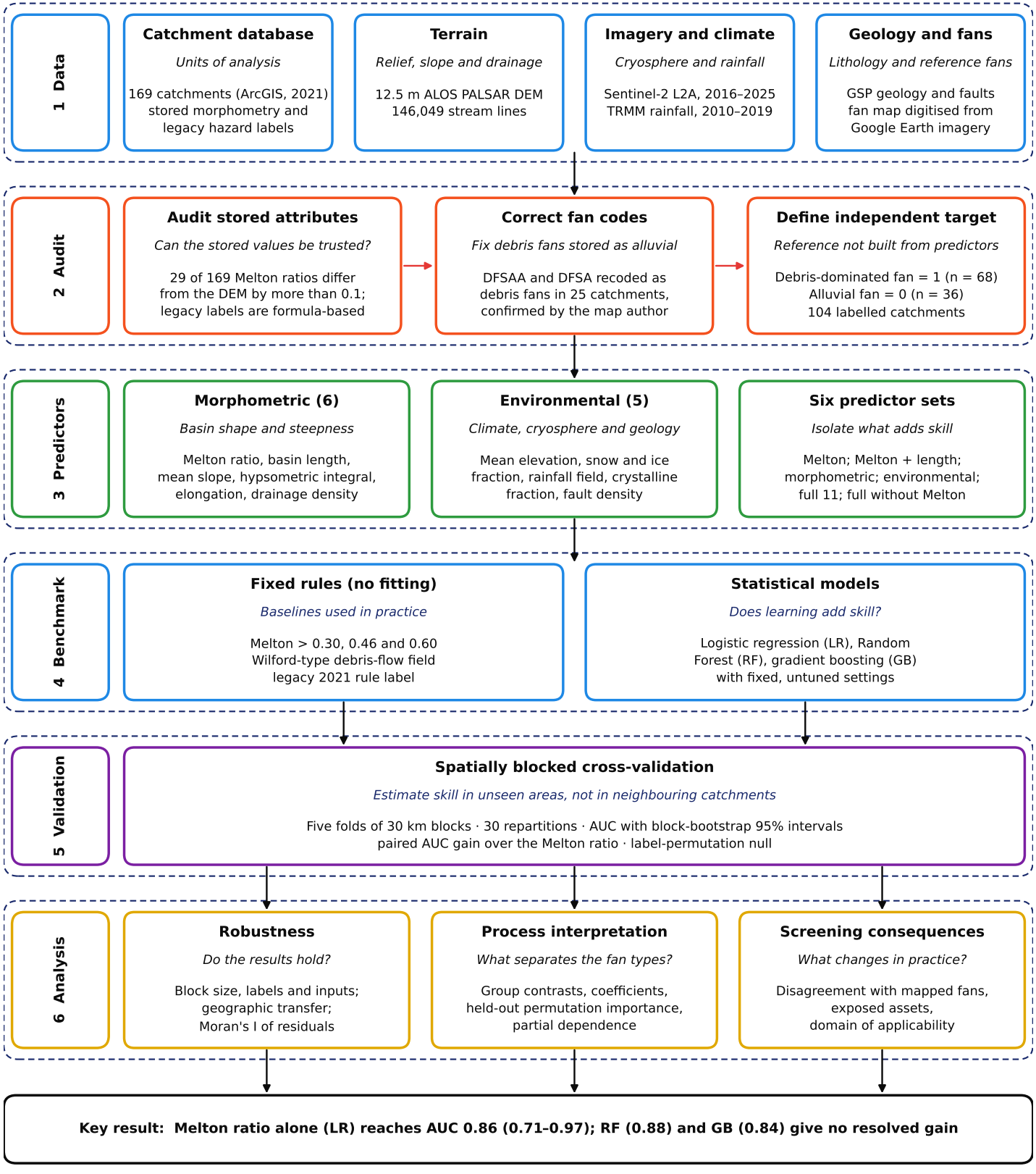

1 Data
Catchment database
Units of analysis
169 catchments (ArcGIS, 2021)
stored morphometry and
legacy hazard labels
Terrain
Relief, slope and drainage
12.5 m ALOS PALSAR DEM
146,049 stream lines
Imagery and climate
Cryosphere and rainfall
Sentinel-2 L2A, 2016–2025
TRMM rainfall, 2010–2019
Geology and fans
Lithology and reference fans
GSP geology and faults
fan map digitised from
Google Earth imagery
2 Audit
Audit stored attributes
Can the stored values be trusted?
29 of 169 Melton ratios differ
from the DEM by more than 0.1;
legacy labels are formula-based
Correct fan codes
Fix debris fans stored as alluvial
DFSAA and DFSA recoded as
debris fans in 25 catchments,
confirmed by the map author
Define independent target
Reference not built from predictors
Debris-dominated fan = 1 (n = 68)
Alluvial fan = 0 (n = 36)
104 labelled catchments
3 Predictors
Morphometric (6)
Basin shape and steepness
Melton ratio, basin length,
mean slope, hypsometric integral,
elongation, drainage density
Environmental (5)
Climate, cryosphere and geology
Mean elevation, snow and ice
fraction, rainfall field, crystalline
fraction, fault density
Six predictor sets
Isolate what adds skill
Melton; Melton + length;
morphometric; environmental;
full 11; full without Melton
4 Benchmark
Fixed rules (no fitting)
Baselines used in practice
Melton > 0.30, 0.46 and 0.60
Wilford-type debris-flow field
legacy 2021 rule label
Statistical models
Does learning add skill?
Logistic regression (LR), Random
Forest (RF), gradient boosting (GB)
with fixed, untuned settings
5 Validation
Spatially blocked cross-validation
Estimate skill in unseen areas, not in neighbouring catchments
Five folds of 30 km blocks · 30 repartitions · AUC with block-bootstrap 95% intervals
paired AUC gain over the Melton ratio · label-permutation null
6 Analysis
Robustness
Do the results hold?
Block size, labels and inputs;
geographic transfer;
Moran's I of residuals
Process interpretation
What separates the fan types?
Group contrasts, coefficients,
held-out permutation importance,
partial dependence
Screening consequences
What changes in practice?
Disagreement with mapped fans,
exposed assets,
domain of applicability
Key result: Melton ratio alone (LR) reaches AUC 0.86 (0.71–0.97); RF (0.88) and GB (0.84) give no resolved gain